

Direct-Margin Scale Recurrence

A Sharp Weyl Law and a Five-Scale Scalar-Alignment Barrier

Prime Dynamics Theory Program

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Abstract

For a threshold τ , define the direct support margin $m_\tau(K) = s_4(K) - \tau s_1(K)$. If, after unitary alignment and a scalar scale $c \geq 0$, the operator error is at most ε , then

$$m_\tau(K') \geq c m_\tau(K) - (1 + \tau)\varepsilon.$$

The coefficients are sharp. We also compute the optimal scalar Chebyshev fit of finite singular profiles. On 96 phase-matched archived pairs, only 26 endpoint fits and 6 four-mode fits retain a positive one-step lower; none of 24 endpoint chains remains positive through all five scales. Thus the direct recurrence is rigorous, but the simplest scalar-alignment mechanism is a poor finite route compared with RH-125.

1 Sharp recurrence

Theorem 1 (Direct support-margin recurrence). *Suppose $\|K' - cUKV\| \leq \varepsilon$ for unitaries U, V and $c \geq 0$. Then for every $\tau \geq 0$,*

$$m_\tau(K') \geq c m_\tau(K) - (1 + \tau)\varepsilon.$$

Proof. Weyl–Mirsky perturbation gives $s_4(K') \geq cs_4(K) - \varepsilon$ and $s_1(K') \leq cs_1(K) + \varepsilon$ [Horn and Johnson, 1991]. Subtraction proves the result. $\square$

The bound is sharp: take a diagonal operator with separated singular values and perturb its leading entry upward by ε and its fourth entry downward by ε .

Corollary 2 (Iteration). *If $m_{n+1} \geq c_n m_n - (1 + \tau)\varepsilon_n$, then*

$$m_N \geq \left(\prod_{j < N} c_j \right) m_0 - (1 + \tau) \sum_{k < N} \varepsilon_k \prod_{j=k+1}^{N-1} c_j.$$

2 Optimal scalar profile fit

For positive profiles s, t , minimize $\varepsilon(c) = \max_j |t_j - cs_j|$ over $c \geq 0$. Feasibility at radius ε is exactly the interval intersection

$$\max_j \frac{t_j - \varepsilon}{s_j} \leq c \leq \min_j \frac{t_j + \varepsilon}{s_j}, \quad c \geq 0.$$

This gives a monotone bisection with a certified minimax residual. We use both the endpoint profile (s_1, s_4) , which contains exactly the margin information, and the stronger four-mode profile.

3 Five-scale barrier

The RH-121 archive supplies 96 adjacent pairs. Every computed recurrence lower is dominated by the target margin. There are zero dominance failures. Yet only 26 endpoint fits retain a positive lower, and only 6 full-profile fits do. Iterating the endpoint recurrence along 24 side-threshold-phase chains leaves zero positive terminal margins; the terminal range is approximately $[-1.40 \times 10^{-2}, -8.45 \times 10^{-11}]$.

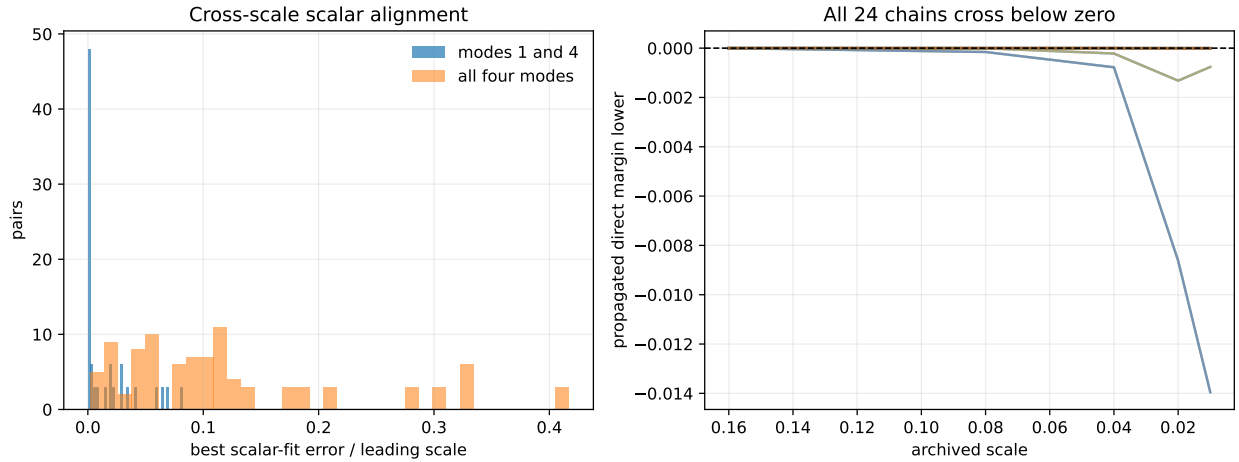


Figure 1: Best scalar-profile errors and propagated direct margins.

This is a finite negative result about scalar alignment, not a proof that every direct physical recurrence is impossible. An operator-valued source law could still succeed. The directional route remains primary, and RH-127 adds the outward guards it needs. No uniform Stage A, Hilbert–Polya, zeta-zero, or Riemann Hypothesis claim is made.

References

Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.